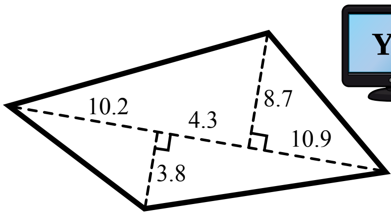
7). This is a sketch of a garden. The measurements are in metres.

a). Calculate the area of the garden.

b). The whole garden is going to be sown with grass seed.
Each square metre needs 65 g of seed.
The seed is bought in 0.75 kg bags.
How many bags does he need? Show all working.

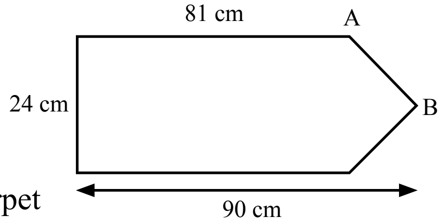


8). A sign-post has one line of symmetry and the dimensions shown.

a). Find the distance AB.

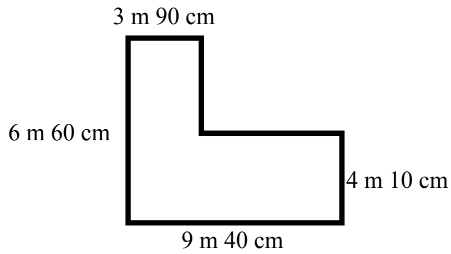
b). Find the perimeter of the sign-post.

c). Find the area of the sign-post.

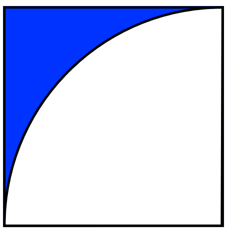
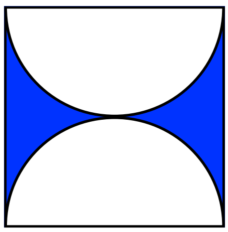
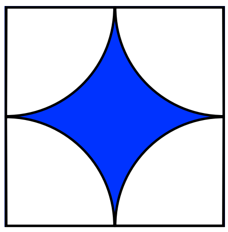
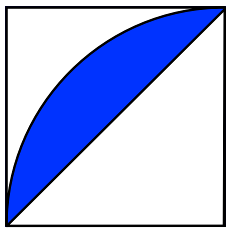


9). A rectangular room, 9.2 m by 6.8 m, is to be covered with square carpet tiles of side length 40 cm.
If each tile costs \$4.90, how much will it cost to cover the room?

10). This 'L' shaped room is to be carpeted.
The dimensions of the room are shown.
The carpet costs \$15.99 per m^2.
How much will it cost to carpet the whole room?



11). Each square below has the same length 32 cm.
For each square find a). the white area b). the shaded area.

i).  ii).  iii).  iv). 

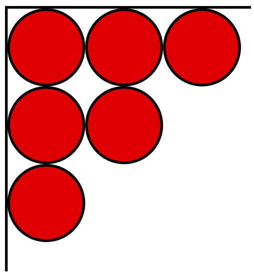
12). Circles of diameter 18 cm are stamped out of a large rectangular piece of metal 2.7 m by 1 m 44 cm as shown in the diagram.

a). Find the area of the metal sheet.

b). Find the area of one of the circles.

c). How many circles can be cut out of the metal sheet?

d). What is the area of waste metal **not** used for circles?



13). A bicycle has a wheel of radius 36 cm.

a). How far will the bicycle travel when the wheel completes
i). 1 revolution, ii). 12 revolutions, iii). 35 revolutions?

b). How many revolutions will the wheel **complete** on a journey of 2 km?

14). The rules for an art competition state the area of the canvas must be exactly 7500 cm^2.

a). John uses a square canvas. What will be the length of one side of the canvas?

b). Lenny uses a **semicircular** canvas. What will be the diameter of the semicircle?

Difficult.

15). a). Jane finds the area of the $^3/_4$ circle to be 1356.5 cm^2.
What is the radius of the circle?

b). Jim has a different $^3/_4$ circle. The **perimeter** of this is 268.4 cm. Find the radius of the circle.